

5 (a)	amplitude of the carrier wave varies	M1	
	in synchrony with displacement of the information/audio signal	A1	[2]
(b) (i)	10 kHz	A1	[1]
(ii)	5 kHz	A1	[1]
(c) (i)	$24 = 10 \lg (P_{\text{MIN}} / \{5.0 \times 10^{-13}\})$	C1	
	$P_{\text{MIN}} = 1.3 (1.26) \times 10^{-10} \text{ W}$	A1	[2]
(ii)	$45 \times 2 = 10 \lg (\{500 \times 10^{-3}\} / P)$		
	$P = 5.0 \times 10^{-10} \text{ (W)}$	M1	
	$P > P_{\text{MIN}}$ so yes	A1	
	<i>or</i>		
	maximum attenuation calculated to be 96 (dB)	(M1)	
	96 dB > 2 × 45 dB so yes	(A1)	
	<i>or</i>		
	maximum length of wire calculated to be 48 (km)	(M1)	
	actual length 45 km < 48 km so yes	(A1)	
	<i>or</i>		
	maximum attenuation per unit length calculated to be 2.2 dB km ⁻¹	(M1)	
	2.2 dB km ⁻¹ > 2.0 dB km ⁻¹ so yes	(A1)	[2]